

DETERMINATION OF COEFFICIENT OF DISCHARGE AND CALIBRATION OF A TRIANGULAR NOTCH

AIM

1. Determination of coefficient of discharge of ' C_d ' of the given triangular notch
2. Plot the graph Q_a Vs H
3. Plot the graph Q_a Vs $H^{5/2}$ find the value of ' C_d ' from the graph
4. Calibrate the notch and determine the actual discharge corresponding to a head of cm

APPARATUS REQUIRED

1. Tank with a Triangular notch
2. Hook gauge
3. Collecting tank with a piezometer.
4. Stop watch

PRINCIPLE

$$\text{Coefficient of discharge, } C_d = \frac{\text{Actual discharge}}{\text{Theoretical discharge}} = \frac{Q_a}{Q_{th}}$$

$$\text{Actual discharge, } Q_a = \frac{A \times R}{t} \text{ cm}^3/\text{sec}$$

Where

A = Area of collecting tank in cm^2

R = Height of water collected in cm

t = Time for collection of ' R ' centimeters of water rise in seconds

Theoretical discharge,

$$Q_{th} = \frac{8}{15} \sqrt{2g} \tan \frac{\phi}{2} H^{5/2} \quad \frac{cm^3}{sec}$$

Where

g = Acceleration due to gravity in cm/s^2

ϕ = Angle of the notch in degrees

H = Head above the sill of the notch = $(h_2 - h_1)$ in cm

h_1 = sill level reading in cm

h_2 = Final hook gauge reading in cm

PROCEDURE

1. The internal plan area of the collecting tank and the angle of the notch are measured.
2. Open the inlet valve and close the bypass valve.
3. Start the pump and water is allowed to rise only up to the sill level of the notch, the supply valve is closed tightly and switch off the pump.
4. The tip of the pointer of the hook gauge is adjusted such that the tip coincides with the free water surface.
5. The sill level of the notch ' h_1 ' is noted from the hook gauge.
6. Start the pump, the supply valve is fully opened, the bypass valve is fully close and the water is allowed to flow through the notch. The tip of the pointer is adjusted to coincide with the water surface.
7. The reading of the hook gauge ' h_2 ' is noted.
8. The out let valve of the collecting tank is tightly closed and the time ' t ' taken for a known rise ' R ' in the collecting tank is noted.
9. Similarly the above procedure is repeated by gradually decreasing the flow by partly opened the bypass valve. (minimum 6 sets)
10. The observations are tabulated and coefficient of discharge is calculated.

OBSERVATIONS AND TABULATION

Sl.No	Hook gauge readings				Time for 'R' Cm rise, t	Actual discharge, Q_a	Theoretical discharge, Q_{th}	Coefficient of discharge, C_d
	Sill level reading, h_1	Final hook gauge reading, b_2	Head over the notch, H $= (h_1 - b_2)$	$H^{5/2}$				
Unit	cm	cm	cm	cm	sec	cm^3/sec	cm^3/sec	
1								
2								
3								
4								
5								
6								

SAMPLE CALCULATION { SET No..... }

Area of collecting tank, A	=	cm ²
Rise of water, R	=	cm
Time for collection, t	=	sec
Actual discharge, Q _a	=	$\frac{A \times R}{t}$
	=	cm ³ /sec
Angle of notch, ϕ	=	degrees
Acceleration due to gravity, g	=	981 cm ² /sec
Sill level reading, h ₁	=	cm
Final hook gauge reading, h ₂	=	cm
Head above the notch, H	=	(h ₂ -h ₁)
	=	cm
Theoretic al discharge, Q _{th}	=	$Q_{th} = \frac{8}{15} \sqrt{2g} \tan \frac{\phi}{2} H^{5/2}$
	=	cm ³ /sec
Coefficient of discharge, Cd	=	$\frac{\text{Actual discharge}}{\text{Theoretical discharge}}$
	=	

CALIBRATION

Actual discharge through the triangular notch is assumed to be governed by the equation,

$$Q_a = KH^n$$

Where

$$Q_a = \text{Actual discharge}$$

$$H = \text{Head over the notch, K and n are Constants}$$

Taking log on both sides

$$\log Q_s = \log K + n \log H$$

Plotting $\log Q_s$ Vs $\log H$ graph, the slope of the line gives the value of n and the intercept on the $\log Q_s$ axis gives $\log K$. Substituting these values of K and n in the equation for discharge $Q_s = KH^n$, the calibration equation can be obtained.

Q_s can be calculated for suitable values of H . The calibration curve can be plotted with H on X-axis and Q_s on Y-axis.

RESULT

Coefficient of discharge of the triangular notch, C_d =

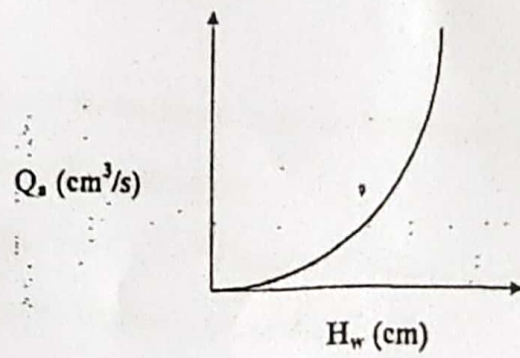
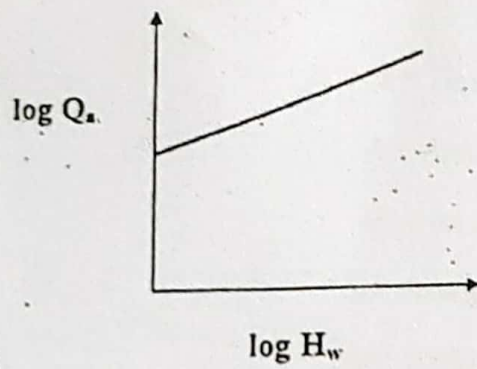
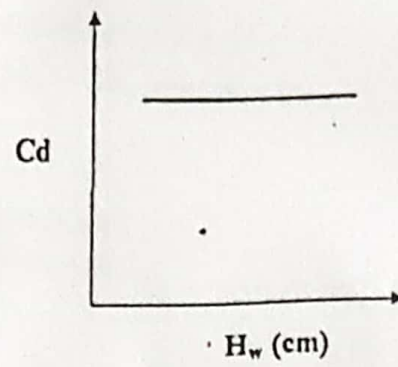
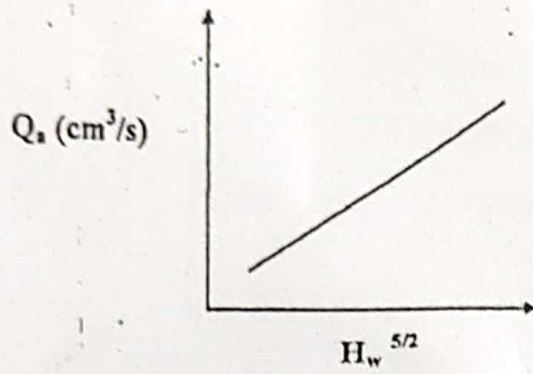
Actual discharge corresponding to a head of flow of cm =

INFERENCE

GRAPHS

1. C_d Vs H
2. Q_s Vs H
3. $\log(Q_s)$ Vs $\log(H)$
4. Calibration graph

Model graphs for Triangular Notch



DETERMINATION OF COEFFICIENT OF DISCHARGE OF AN ORIFICE BY TIME OF EMPTYING METHOD.

AIM:-

To determine the coefficient of discharge of (C_d) of the orifice by varying head method and plot the graph. ($\sqrt{h_1} - \sqrt{h_2}$) v/s 'T'. Taking 'T' on y-axis.

APPARATUS USED

1. Orifice fitted to an orifice tank.
2. Orifice tank with a piezometer.
3. Stop watch.

PRINCIPLE

The coefficient of discharge is found out using the expression.

$$\text{Time } T = \frac{2A (\sqrt{h_1} - \sqrt{h_2})}{C_d a \sqrt{2g}}$$

$$C_d = \frac{2A(\sqrt{h_1} - \sqrt{h_2})}{T a \sqrt{2g}}$$

Where

T = Time required for the liquid to descend from head (h_1) to head (h_2) in sec.

A = Area of the orifice tank in cm^2

a = Area of orifice = $\pi/4 d^2$ (d = diameter of orifice) in cm^2

g = Acceleration due to gravity in cm/s^2

h_1 = Initial head in the orifice tank in cm

h_2 = Final head in the orifice tank in cm

C_d = Coefficient of discharge

PROCEDURE

1. The internal plan dimensions of the orifice tank and the diameter of the orifice are measured.
2. Start the motor and by adjusting the bypass valve, the water is allowed into the orifice tank. So that the head above the centre of the orifice is (h_1).
3. The supply valve is completely closed (switch off the machine) and the water level in the orifice tank is allowed to descend.

OBSERVATIONS AND TABULATION

Sl.No	h_1	h_2	$\sqrt{h_1}$	$\sqrt{h_2}$	$\sqrt{h_1 - h_2}$	Time 'T'	C_d
1							
2							
3							
4							
5							
6							
	Diameter of Orifice = cm						
	Area of Orifice tank = cm ²						

4. When the descending water crosses the selected head (h_1) the stop watch is started and the time "T" taken for the water to cross the head (h_2) is noted.
5. The above procedure is repeated for different values of h_1 and h_2 and the time taken in each case is observed. (Minimum 6 Sets)
6. The observations are tabulated and the coefficient of discharge is calculated.
- 7.

SAMPLE CALCULATION {SET No:.....}

T = Time required for the liquid to descend from head (h_1) to head (h_2) in sec.

A = Area of the orifice tank in cm^2
=

a = Area of orifice = $\pi/4 d^2$ (d= diameter of orifice) in cm^2

g = Acceleration due to gravity in cm/s^2

h_1 = cm

h_2 = cm

C_d = Coefficient of discharge

$$C_d = \frac{2A(\sqrt{h_1} - \sqrt{h_2})}{T a \sqrt{2g}}$$

=

RESULT

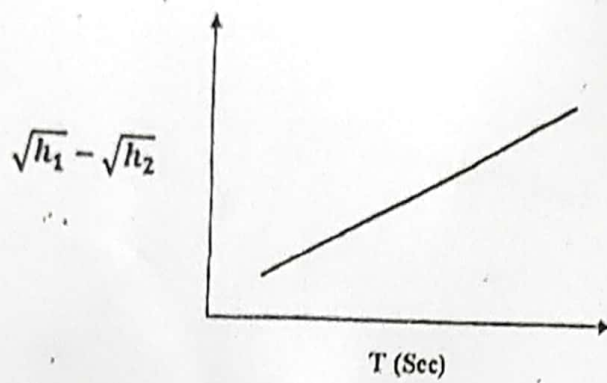
Coefficient of discharge, C_d =

INFERENCE

GRAPH

1. T Vs $\sqrt{h_1} - \sqrt{h_2}$

COEFFICIENT OF DISCHARGE OF AN ORIFICE BY TIME OF
EMPTYING METHOD



DETERMINATION OF METACENTRIC HEIGHT OF A

FLOATING BODY

AIM:-

To determine the metacentric height and radius of Gyration of the given floating body and plot the graph between metacentric height Vs distance

APPARATUS

1. A water tank filled up to $3/4^{\text{th}}$ with water.
2. A ship model with arrangement to measure the tilt due to the displacement of small weight.
3. Stop watch.

PRINCIPLE

The ship model is tilted for an angle Φ from its horizontal position by a small extra weight 'w' on one side of the deck of the ship model at a distance of 'x' from the centre of the duck.

The metacentric height, GM (distance between the centre of gravity and metacentric) is calculated as

$$GM = \frac{wx}{(W+w) \tan \Phi}$$

Where

w = Small extra weight.

W = Weight of the ship model.

x = Distance of that weight 'w' mounted from the centre measured along the duck in cm.

Φ = Angle measured in degrees.

OBSERVATIONS AND TABULATION						
Sl. No	Weight 'w'	Distance of weight		Angle of head		Metacentric height
		x_L	x_R	Φ_L	Φ_R	
Unit	Grams	cm	cm	Angle	Angle	
1						
2						
3						
4						
5						
6						
RADIUS OF GYRATION						
Sl.No.	No. of oscillations	Time for 'n' oscillation 't'	Time for one oscillations $T = t/n$	Mean time of oscillation'	Radius of Gyration .K	
Unit	Nos	s	s	s		
1						
2						
3						
4						
5						
6						

Radius of Gyration (K)

When a rotating body behaves as if all its mass were concentrated in a ring at a distance of 'K' from the axis of rotation, that distance is called the Radius of Gyration of the body.

Radius of Gyration,

$$K = \frac{T\sqrt{g GM}}{2\pi}$$

Where

T = Time for one oscillation in seconds

GM = Metacentric height in cm

PROCEDURE:-

1. Place the ship model in the tank having water for about $3/4^{\text{th}}$ height. Bring the float into a horizontal position by rotating the weight at the ends by adjusting the nut inwards or outwards so that the plumb line passes through zero mark of the tilt measuring device.
2. Place a small weight 'w' on one side (say left) of the float deck at a distance of x_L cm from the centre. Note the angle Φ_R (right) shown by the plumb line. Remove the weight from the left side and put it on the other side (right). Note the tilt shown by the plumb line Φ_L (Left).
3. Repeat the steps for two or more distances from the centre and one more small weight w_2 in order to get 6 sets of readings.
4. Remove the weight and allow the ship to oscillate. Note the time in seconds taken for 10 cycles of oscillations. Repeat the above steps for two more different cycles and then calculate mean time of oscillation. Tabulate the readings.

RESULT:-

INFERENCE:-

VARIFICATION OF BERNOULLI'S THEOREM

AIM :-

1. To calculate the total energy at different points.
2. To plot the graphs, Energy Vs Distance, Head Vs Distance and Velocity Vs Distance.

PRINCIPLE

$$\text{Total energy, } E = \frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$$

Where,

E = Total energy

$\frac{P_1}{\rho g}$ = Pressure energy per unit weight of fluid or pressure head at point 1.

$\frac{V_1^2}{2g}$ = Kinetic energy per unit weight or Kinetic head at point 1.

Z_1 = Potential energy per unit weight of potential head at point 1.

$\frac{P_2}{\rho g}$ = Pressure head at point 2.

$\frac{V_2^2}{2g}$ = Kinetic head at point 2.

Z_2 = Potential head at point 2.

P_1 = Pressure of fluid at point 1.

P_2 = Pressure of fluid at point 2.

OBSERVATIONS AND TABULATION									
SI No	Time for R cm rise of water	Head 'h' at piezometer tube no							
		1	2	3	4	5	6	7	
1									
2									
3									
4	Discharge, $Q \text{ cm}^3/\text{s}$								
5	Diameter of this station in cm								
6	Area, $a \text{ cm}^2$								

Tube No	1	2	3	4	5	6	7
Velocity (Q/a)							
Velocity head, cm							
Pressure head, cm							
Datum head, cm							
Total Head, cm							

R = Rise of water = cm

V_1 = Velocity of fluid at point 1.

V_2 = Velocity of fluid at point 2.

ρ = Density of fluid.

g = Acceleration due to gravity.

$h = \frac{P}{\rho g}$ (Pressure head)

Formula,

$$\text{Total energy, } E = \frac{P}{\rho g} + \frac{V^2}{2g} + Z$$

$$\text{Velocity of flow, } V = \frac{\text{Actual discharge}}{\text{Area of cross section}} = \frac{Q}{a} \text{ cm/sec.}$$

$$\text{Actual discharge, } Q = \frac{A R}{t} \frac{\text{cm}^3}{\text{Sec}}$$

Where,

A = Area of collecting tank in m^2

R = Rise of water in m

T = Time for 'R' rise of water in sec

Z = Datum head

SAMPLE CALCULATION {SET No:.....}

$$\text{Actual Discharge} = \frac{A R}{t} = \frac{\text{cm}^3}{\text{Sec}}$$

$$\text{Area of cross-section of pipe, } a = \text{cm}^2$$

$$\text{Velocity of flow, } v = \frac{Q}{a} \frac{\text{cm}}{\text{sec}}$$

$$\text{Velocity Head} = \frac{v^2}{2g} \text{ cm}$$

$$\text{Pressure Head} = \text{cm}$$

$$\text{Datum Head} = 0 \text{ or Constant}$$

Total Energy = Pressure Head + Velocity Head + Datum Head

PROCEDURE

1. Start the pump and allow the water to the tank up to a particular head.
2. Regulate the flow of water through the test section with the help of gate valve.
3. Note the pressure head 'h' at each test section. (7 Tubes)
4. The outlet valve of collecting tank is fully closed and note the time 't' for 'R' rise of water.
5. Regulate the flow by using the gate valve, change the pressure head and observe the above readings.
6. The above procedure is repeated by gradually increasing /decreasing the flow.
7. Note the diameter of the each section and also note the distances from the reference points.

RESULT:-

1. The energy at various points is more or less the same.
2. Graphs are plotted.

INFERENCE

DETERMINATION OF COEFFICIENT OF DISCHARGE AND CALIBRATION OF ORIFICE METER

AIM:-

1. To determine the coefficient of discharge of the given Orifice meter and plot the graph Q_a Vs H_m
2. To calibrate the Orifice meter.

APPARATUS REQUIRED

1. Orifice meter fitted with a pipe.
2. Differential 'U' tube mercury column manometer.
3. Collecting tank fitted with a piezometer.
4. Stop watch.

PRINCIPLE

Theoretical discharge Q_{th} through the orifice meter is given by,

$$Q_{th} = \frac{a_1 a_2 \sqrt{2 g H_w}}{\sqrt{(a_1^2 - a_2^2)}} \quad \text{cm}^3/\text{s}$$

Where,

a_1 = Area of the pipe = $\frac{\pi}{4} d_1^2$ in cm^2 , where d_1 the diameter of the pipe in cm.

a_2 = Area of the orifice = $\frac{\pi}{4} d_2^2$ in cm^2 , where d_2 the diameter of the orifice in cm.

H_w = Manometer head in cm of water.

$$= H_m \frac{(S_m - S_l)}{S_l}$$

H_m = Manometer head in cm of Hg = $h_2 - h_1$

h_1 = left limb manometer reading in cm of Hg.

h_2 = Right limb manometer reading in cm of Hg.

S_m = Specific gravity of manometric liquid (Hg) = 13.6

S_l = Specific gravity of flowing liquid (water) = 1

$$\text{Actual discharge, } Q_a = \frac{A \times R}{t} \text{ cm}^3/\text{s}$$

Where

A = Area of collecting tank in cm^2 .

R = Rise of water in cm.

t = Time for the 'R' rise of water in seconds.

Coefficient of discharge through the orifice,

$$C_d = \frac{\text{Actual discharge}}{\text{Theoretical discharge}} = \frac{Q_a}{Q_{th}}$$

PROCEDURE

1. The diameter of the orifice and the diameter of the pipe are noted.
2. The internal plan dimensions of the collecting tank are measured.
3. Start the pump and keep the outlet valve fully open
4. Then the water is allowed to flow through the orifice meter.
5. Note the manometer readings.
6. The out let valve of the collecting tank is closed and the time 't' for 'R' rise of water in the collecting tank is noted.
7. The above procedure is repeated by gradually decreasing the flow by partly opening the bypass valve.(6 sets)
8. The observations are tabulated and the coefficient of discharge is calculated.

OBSERVATIONS AND TABULATION							
S.No	Manometer readings			Time for 'R' cm rise, t	Actual discharge Q_a cm^3/s	Theoretical discharge Q_{th} cm^3/s	Coefficient of discharge, C_d
	Left limb h_1 cm of Hg	Right limb h_2 cm of Hg	$H_m = (h_2 - h_1)$ cm of Hg cm of water $H_w = H_m \times 12.6$				
Unit	cm of Hg	cm of Hg	cm of Hg	sec	cm^3/s	cm^3/s	
1							
2							
3							
4							
5							
6							

SAMPLE CALCULATION {SET No.....}

Area of collecting tank, A	=	cm ²
Rise of water, R	=	cm
Time for collection, t	=	sec
Actual discharge, Q _a	=	cm ³ /s
Diameter of inlet, d ₁	=	cm
Area of inlet, a ₁	=	cm ²
Diameter of throat, d ₂	=	cm
Area of throat, a ₂	=	cm ²
Acceleration due to gravity, g	=	cm/s ²
Left limb manometer reading, h ₁	=	cm
Right limb manometer reading, h ₂	=	cm
Specific gravity of manometric liquid, S _m	=	13.6
Specific gravity of flowing liquid, S ₁	=	1
Manometer head, H _w	=	cm of water
Theoretical discharge, Q _{th}	=	cm ³ /s
Coefficient of discharge, C _d	=	

CALIBRATION

Actual discharge through the orifice meter is assumed to be governed by the equation,

$$Q_a = KH_w^n$$

Where

Q_a = Actual discharge

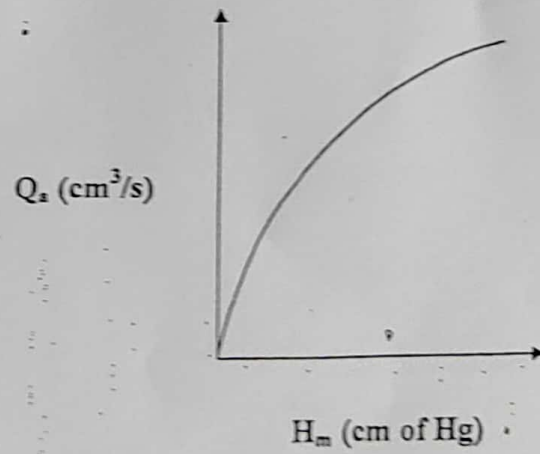
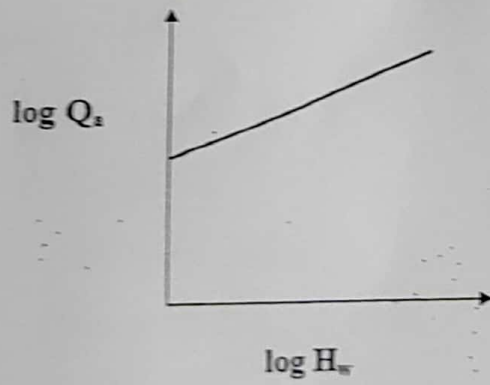
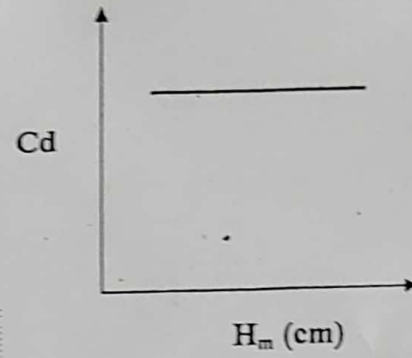
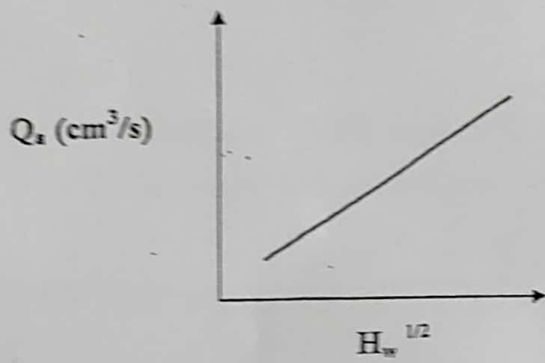
H_w = Pressure difference between the inlet and the throat
(cm of water)

K and n are constants

Taking log on both sides

$$\log Q_a = \log K + n \log H_w$$

Model graphs for Orificemeter



Plotting $\log Q_a$ Vs $\log H_w$ graph the slope of the line gives the value of n and the intercept on the $\log Q_a$ axis gives the $\log K$, Substituting these values of K and n in the equation for discharge $Q_a = K H_w^n$, the calibration equation can be obtained.

Q_a can be calculated for suitable values of H_m (converted to H_w). The calibration curve can be plotted with H_m on X-axis and Q_a on Y-axis.

RESULT

1. Coefficient of discharge of the Orifice meter is, $C_d =$
2. Flow rate of actual discharge to the head of 8 cm of Hg = cm^3/sec

INFERENCE

GRAPHS

1. C_d Vs H_m .
2. Q_a Vs H_m .
3. $\log Q_a$ Vs $\log H_w$.
4. Calibration Graph, Q_a Vs H_m